

Seven Stages: From Text Features to Transformers

A consistent learning map with examples and plain-language explanations.

Stage	Topic	Examples	Simple explanation and use case
1	Traditional machine learning for text	TF-IDF; Naive Bayes; Logistic Regression; SVM	Turn text into counts or TF-IDF numbers, then train an algorithm to predict a category. Example: classify a support message as a card, transfer, or loan issue.
2	Word embeddings	Word2Vec; GloVe; FastText	Represent words with learned vectors (lists of numbers). Related usage can produce related vectors. These vectors are inputs to other models; they are not complete complaint classifiers by themselves.
3	Convolutional neural networks (CNN)	2D CNN for images; 1D CNN for text	Move learned filters across nearby positions to detect patterns. A 2D CNN can find shapes in a document image; a 1D text CNN can detect a phrase such as “charged twice.”
4	Recurrent sequence networks	RNN; LSTM; GRU	Read an ordered sequence step by step and carry a state forward. LSTM and GRU use gates to manage information. Example: read a customer message in word order before predicting its intent.
5	Transformer encoder	BERT; RoBERTa	Use attention so each input token can draw on other input tokens. Produce context-aware representations for classification, information extraction, and search. Example: identify the issue in a complaint.
6	Transformer decoder	GPT; Llama	Generate text one token at a time, using available earlier tokens. A causal mask blocks future output tokens. Example: draft a response to a customer message for human review.
7	Transformer encoder-decoder	T5; BART	The encoder reads and represents the input; the decoder writes an output while attending to that input. Example: turn a long complaint into a short support summary.

Bridge to transformers: Learn attention between Stages 4 and 5. Then study tokenization, position information, self-attention, queries/keys/values (Q/K/V), and multi-head attention before exploring the three transformer architectures.

How to read this table: These are teaching stages, not mandatory steps in one model. Stage 2 describes a representation; Stages 3-7 describe neural network architectures.